3. Midpoint

It is obtained by dividing the sum of lower limit and upper limit by 2.

For example in the class 10 – 20, the midpoint is $\frac{10+20}{2}=15$

4. Class interval

It is the difference between the upper and lower limit of a class.

For example:

In the class 10 – 20,

The class interval is 20 – 10 = 10.

5. Tally Marks

Tally marks usually come in group of 5. For every observation, we put vertical bars called tally marks. After 4 bars we put a cross stroke for 5th observation. It helps in easy counting of the frequencies.

6. Class Frequency

The number of observations included in each class is the class frequency or sum of tally marks of that particular class.

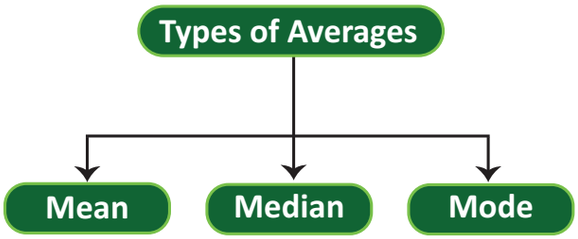
7. Total Frequency

It is an aggregate of all class frequencies. It is the sum of all observations.

13.6 Statistical techniques - Measures of central tendency

One of the most important objectives of statistical analysis is to get one single value that describes the characteristics of the entire data. A common word used for measures of central tendency is average. For example, if we collect the heights of 5000 students of a college, we will have 5000 figures. The persons mind is not capable of registering all the data at a time. So we need a single figure which represents the whole data. This single figure is known as average.

Since an average represents the entire data, its value lies somewhere in between the two extremes i.e. the largest observation and the smallest observation. For this reason the average is frequently referred as measure of central tendency.



Mean

Mean is defined as the value obtained by dividing sum of values of all the given items by the number of items. It is also called as arithmetic mean.

Formula

$$\text{Mean} = \frac{\text{Sum of the values of all items}}{\text{Total number of items}}$$

Mathematically it can be shown as

$$\bar{X} = \frac{X_1 + X_2 + X_3 + \ldots\ldots X_n}{N}$$

$$\bar{X} = \frac{\Sigma X}{N}$$

$\bar{X}$ = Mean

ΣX = Sum of the Values of all items

N = Total number of items

Calculation of Mean

Example 13.1

The following are the monthly average temperature data of Cuddalore District. Find the mean annual temperature.

Cuddalore

Temperature^0 C	Jan	Feb	Mar	April	May	June	July	Aug	Sep	Oct	Nov	Dec
	25.3	26.58	28.46	30.61	31.32	30.75	30	29.34	29.03	27.89	26.45	25.36

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